

ELITE MATHEMATICS

Pearson Edexcel International A Level

Mechanics 1

Adaptive Classified

Chapter 04 · Constant Acceleration in 1D

Questions

Student book · questions with writing space

- 52 pages
- Past papers up to June 2026

MECHANICS 1 | TOPIC 04

Constant Acceleration in 1D

Your classified route

17

QUESTIONS

162

MARKS

5

FAMILIES

11

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Read the diagram, choose the model, then use the working space to complete the mechanics.

TOPIC 04 | FAMILY 01

Straight-line constant-acceleration SUVAT

WHAT THIS FAMILY ASKS

Choose and apply a scalar constant-acceleration relation in a fixed sign convention.

VARIANT ROUTE

V01 Choose a SUVAT relation

V02 Distance in the n th second or sub-interval

V03 Recover a parameter from average speed

4 classified questions

37 total marks

5. The points O , A , B and C are on a rough horizontal surface where $OABC$ is a straight line.

A particle is projected horizontally from O and slides across the surface, passing through A , B and C in that order.

The particle passes through

- A with speed $U \text{ m s}^{-1}$
- B with speed $\frac{U}{2} \text{ m s}^{-1}$

The coefficient of friction between the particle and the surface is $\frac{1}{7}$

It takes 0.75 s for the particle to move from A to B .

- (a) Show that $U = 2.1$ to 2 significant figures.

(6)

DR ESAM AHMED

The particle passes through

- A with speed $U \text{ m s}^{-1}$
- B with speed $\frac{U}{2} \text{ m s}^{-1}$

The coefficient of friction between the particle and the surface is $\frac{1}{7}$

$U = 2.1$ to 2 significant figures.

The particle passes through C with speed $\frac{U}{3} \text{ m s}^{-1}$

(b) Find the time taken to move from B to C .

(3)

DR ESAM AHMED

5. The points O , A , B and C are on a rough horizontal surface where $OABC$ is a straight line.

A particle is projected horizontally from O and slides across the surface, passing through

The particle passes through

- A with speed $U \text{ m s}^{-1}$
- B with speed $\frac{U}{2} \text{ m s}^{-1}$

The coefficient of friction between the particle and the surface is $\frac{1}{7}$

It takes 0.75 s for the particle to move from A to B .

$U = 2.1$ to 2 significant figures.

The particle passes through C with speed $\frac{U}{3} \text{ m s}^{-1}$

(c) Find the distance AC .

(3)

DR ESAM AHMED

2. A car moves along a straight horizontal road with constant acceleration $a \text{ ms}^{-2}$ where $a > 0$

The car is modelled as a particle.

At time $t = 0$, the car passes point A and is moving with speed $u \text{ ms}^{-1}$

In the first three seconds after passing A the car travels 20 m.

In the fourth second after passing A the car travels 10 m.

The speed of the car as it passes point B is 20 ms^{-1}

Find the time taken for the car to travel from A to B .

(8)

DR ESAM AHMED

2. A motorbike is moving with constant acceleration along a straight horizontal road.

The motorbike passes a point P and 10 seconds later passes a point Q .

The speed of the motorbike as it passes Q is 28 m s^{-1}

Given that $PQ = 220 \text{ m}$,

- (a) find the acceleration of the motorbike,

(3)

DR ESAM AHMED

2. A motorbike is moving with constant acceleration along a straight horizontal road.

The motorbike passes a point P and 10 seconds later passes a point Q .

The speed of the motorbike as it passes Q is 28 m s^{-1}

Given that $PQ = 220 \text{ m}$,

- (b) find the distance travelled by the motorbike during the fifth second after passing P (4)

DR ESAM AHMED

3. A van travels with constant acceleration along a straight horizontal road.

The van passes a point A with speed $u \text{ m s}^{-1}$ and 20 seconds later passes a point B with speed 28 m s^{-1}

The distance AB is 400 m.

- (a) Show that $u = 12$

(2)

DR ESAM AHMED

3. A van travels with constant acceleration along a straight horizontal road.

The van passes a point A with speed $u \text{ m s}^{-1}$ and 20 seconds later passes a point B with speed 28 m s^{-1}

The distance AB is 400 m.

- (b) Find the time taken for the van to travel from A to the midpoint of AB .

(5)

DR ESAM AHMED

3. A van travels with constant acceleration along a straight horizontal road.

The van passes a point A with speed $u \text{ m s}^{-1}$ and 20 seconds later passes a point B with speed 28 m s^{-1}

The distance AB is 400 m.

The van has mass 1200 kg.

During its motion the van experiences a constant resistive force of magnitude 260 N

- (c) Find the magnitude of the driving force exerted by the engine of the van as it travels from A to B .

(3)

DR ESAM AHMED

TOPIC 04 | FAMILY 02

Vertical motion under gravity

WHAT THIS FAMILY ASKS

Apply a single upward/downward sign convention with acceleration of magnitude g .

VARIANT ROUTE

V01 Time or speed at ground impact

V02 Duration above a level

5 classified questions

43 total marks

1. A small stone is projected vertically upwards with speed 20 m s^{-1} from a point O which is 5 m above horizontal ground. The stone is modelled as a particle moving freely under gravity.

Find

- (a) the speed of the stone at the instant when it is 2 m above the ground,

(2)

DR ESLAM AHMED

1. A small stone is projected vertically upwards with speed 20 m s^{-1} from a point O which is 5 m above horizontal ground. The stone is modelled as a particle moving freely under gravity.

Find

- (b) the total time between the instant when the stone is projected from O and the instant when it first strikes the ground.

(4)

DR ESAM AHMED

4. At time $t = 0$, a small ball is projected vertically upwards from a point A which is 24.5 m above the ground. The ball first comes to instantaneous rest at the point B , where $AB = 19.6$ m and first hits the ground at time $t = T$ seconds.

The ball is modelled as a particle moving freely under gravity.

- (a) Find the value of T .

(6)

DR ESAM AHMED

4. At time $t = 0$, a small ball is projected vertically upwards from a point A which is 24.5 m above the ground. The ball first comes to instantaneous rest at the point B , where $AB = 19.6$ m and first hits the ground at time $t = T$ seconds.

The ball is modelled as a particle moving freely under gravity.

- (b) Sketch a speed-time graph for the motion of the ball from $t = 0$ to $t = T$ seconds.

(No further calculations are needed in order to draw this sketch.)

(2)

DR ESAM AHMED

3. A particle is projected vertically upwards with speed $U \text{ m s}^{-1}$ from a point A .

The point A is 12 m vertically above the point B .

Point B is on horizontal ground.

The particle moves freely under gravity until it hits the ground at B .

The time taken for the particle to travel from A to B is 4 seconds.

- (a) Find the value of U .

(3)

DR ESAM AHMED

3. A particle is projected vertically upwards with speed $U \text{ m s}^{-1}$ from a point A .

The point A is 12 m vertically above the point B .

Point B is on horizontal ground.

The particle moves freely under gravity until it hits the ground at B .

The time taken for the particle to travel from A to B is 4 seconds.

- (b) Find the speed of the particle as it hits the ground at B .

(3)

DR ESAM AHMED

3. A particle is projected vertically upwards with speed $U \text{ m s}^{-1}$ from a point A .

The point A is 12 m vertically above the point B .

Point B is on horizontal ground.

The particle moves freely under gravity until it hits the ground at B .

The time taken for the particle to travel from A to B is 4 seconds.

- (c) Sketch a speed-time graph for the motion of the particle from the instant it leaves A to the instant it reaches B . (No further calculations are required.)

(2)

DR ESAM AHMED

5. A small ball is projected vertically upwards with speed 29.4 m s^{-1} from a point A which is 19.6 m above horizontal ground.

The ball is modelled as a particle moving freely under gravity until it hits the ground. It is assumed that the ball does not rebound.

- (a) Find the distance travelled by the ball while its speed is less than 14.7 m s^{-1}

(3)

DR ESLAM AHMED

5. A small ball is projected vertically upwards with speed 29.4 m s^{-1} from a point A which is 19.6 m above horizontal ground.

The ball is modelled as a particle moving freely under gravity until it hits the ground. It is assumed that the ball does not rebound.

- (b) Find the time for which the ball is moving with a speed of more than 29.4 m s^{-1}

(3)

DR ESAM AHMED

5. A small ball is projected vertically upwards with speed 29.4 m s^{-1} from a point A which is 19.6 m above horizontal ground.

The ball is modelled as a particle moving freely under gravity until it hits the ground. It is assumed that the ball does not rebound.

- (c) Sketch a speed-time graph for the motion of the ball from the instant when it is projected from A to the instant when it hits the ground. Show clearly where your graph meets the axes.

(3)

DR ESAM AHMED

6. A particle is projected vertically upwards from a point A with speed 24 m s^{-1}

The point A is 2.5 m vertically above the point B .

Point B lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at B with speed $V \text{ m s}^{-1}$

After hitting the ground the particle does not rebound.

- (a) Find the value of V .

(3)

DR ESAM AHMED

6. A particle is projected vertically upwards from a point A with speed 24 m s^{-1}

The point A is 2.5 m vertically above the point B .

Point B lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at B with speed $V \text{ m s}^{-1}$

After hitting the ground the particle does not rebound.

- (b) Find the time taken for the particle to reach B .

(3)

DR ESAM AHMED

6. A particle is projected vertically upwards from a point A with speed 24 m s^{-1}

The point A is 2.5 m vertically above the point B .

Point B lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at B with speed $V \text{ m s}^{-1}$

The point C is 10 m vertically above A .

- (c) Find the length of time for which the particle is above C .

(4)

DR ESAM AHMED

6. A particle is projected vertically upwards from a point A with speed 24 m s^{-1}

The point A is 2.5 m vertically above the point B .

Point B lies on horizontal ground.

The particle moves freely under gravity until it hits the ground at B with speed $V \text{ m s}^{-1}$

After hitting the ground the particle does not rebound.

- (d) Sketch a speed-time graph for the motion of the particle from projection to the instant that it reaches B . (No further calculations are required.)

(2)

DR ESAM AHMED

TOPIC 04 | FAMILY 03

Piecewise one-dimensional motion

WHAT THIS FAMILY ASKS

Join constant-acceleration stages when force, acceleration, parachute state or model changes.

VARIANT ROUTE

V01 Acceleration or force switches value

V02 Compare alternative piecewise motion models

3 classified questions

33 total marks

5.

**Figure 2**

Three points P , Q and R are on a horizontal road where PQR is a straight line.

The point Q is between P and R , with $PQ = 6x$ metres and $QR = 5x$ metres, as shown in Figure 2.

A vehicle moves along the road from P to Q with constant acceleration.

The vehicle is modelled as a particle.

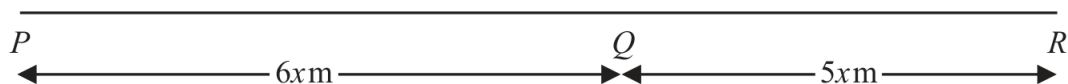
At time $t = 0$, the vehicle passes P with speed $u \text{ m s}^{-1}$

At time $t = 12 \text{ s}$, the vehicle passes Q with speed $2u \text{ m s}^{-1}$

Using the model,

(a) show that $x = 3u$

(2)



A vehicle moves along the road from P to Q with constant acceleration.

At time $t = 0$, the vehicle passes P with speed $u \text{ ms}^{-1}$

At time $t = 12 \text{ s}$, the vehicle passes Q with speed $2u \text{ ms}^{-1}$

$$t \ x = 3u$$

As the vehicle passes Q , the acceleration of the vehicle changes instantaneously to 1.5 ms^{-2}

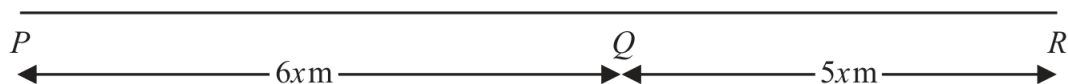
The vehicle continues to move with a constant acceleration of 1.5 ms^{-2} and passes R with speed $3u \text{ ms}^{-1}$

Using the model,

(b) find the value of u ,

(3)

DR ESAM AHMED



A vehicle moves along the road from P to Q with constant acceleration.

At time $t = 0$, the vehicle passes P with speed $u \text{ m s}^{-1}$

At time $t = 12 \text{ s}$, the vehicle passes Q with speed $2u \text{ m s}^{-1}$

$$t \ x = 3u$$

As the vehicle passes Q , the acceleration of the vehicle changes instantaneously to 1.5 m s^{-2}

(c) find the distance travelled by the vehicle during the first 14 seconds after passing P

(4)

DR ESAM AHMED

5. A parachute is used to deliver a box of supplies. The parachute is attached to the box.
- the parachute and box are dropped from rest from a helicopter that is hovering at a height of 520 m above the ground
 - the parachute and box fall vertically and freely under gravity for 5 seconds, then the parachute opens
 - from the instant the parachute opens, it provides a resistance to motion of magnitude 3200 N
 - the parachute and box continue to fall vertically downwards after the parachute opens
 - the parachute and box are modelled throughout the motion as a particle P of mass 250 kg

(a) Find the distance fallen by P in the first 5 seconds.

(2)

DR ESAM AHMED

5. A parachute is used to deliver a box of supplies. The parachute is attached to the box.

- the parachute and box are dropped from rest from a helicopter that is hovering at a height of 520 m above the ground
- the parachute and box fall vertically and freely under gravity for 5 seconds, then the parachute opens
- from the instant the parachute opens, it provides a resistance to motion of magnitude 3200 N
- the parachute and box continue to fall vertically downwards after the parachute opens
- the parachute and box are modelled throughout the motion as a particle P of mass 250 kg

(b) Find the speed with which P lands on the ground.

(7)

DR ESAM AHMED

5. A parachute is used to deliver a box of supplies. The parachute is attached to the box.

- the parachute and box are dropped from rest from a helicopter that is hovering at a height of 520 m above the ground
- the parachute and box fall vertically and freely under gravity for 5 seconds, then the parachute opens
- from the instant the parachute opens, it provides a resistance to motion of magnitude 3200 N
- the parachute and box continue to fall vertically downwards after the parachute opens
- the parachute and box are modelled throughout the motion as a particle P of mass 250 kg

(c) Find the total time from the instant when P is dropped from the helicopter to the instant when P lands on the ground.

(3)

DR ESAM AHMED

5. A parachute is used to deliver a box of supplies. The parachute is attached to the box.

- the parachute and box are dropped from rest from a helicopter that is hovering at a height of 520 m above the ground
- the parachute and box fall vertically and freely under gravity for 5 seconds, then the parachute opens
- from the instant the parachute opens, it provides a resistance to motion of magnitude 3200 N
- the parachute and box continue to fall vertically downwards after the parachute opens
- the parachute and box are modelled throughout the motion as a particle P of mass 250 kg

(d) Sketch a speed-time graph for the motion of P from the instant when P is dropped from the helicopter to the instant when P lands on the ground.

(2)

DR ESAM AHMED

3. A car is moving at a constant speed of 25 m s^{-1} along a straight horizontal road.

The car is modelled as a particle.

At time $t = 0$, the car is at the point A and the driver sees a road sign 48 m ahead.

Let t seconds be the time that elapses after the car passes A .

In a **first** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until the car reaches the road sign.

- (a) Use this first model to find the speed of the car as it reaches the sign.

(2)

DR ESAM AHMED

3. A car is moving at a constant speed of 25 m s^{-1} along a straight horizontal road.

The car is modelled as a particle.

At time $t = 0$, the car is at the point A and the driver sees a road sign 48 m ahead.

Let t seconds be the time that elapses after the car passes A .

The road sign indicates that the speed limit immediately after the sign is 13 m s^{-1} .

In a **second** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until it reaches a speed of 13 m s^{-1} . The car then maintains this speed until it reaches the road sign.

- (b) Use this second model to find the value of t at which the car reaches the sign.

(4)

DR ESAM AHMED

3. A car is moving at a constant speed of 25 m s^{-1} along a straight horizontal road.

The car is modelled as a particle.

At time $t = 0$, the car is at the point A and the driver sees a road sign 48 m ahead.

Let t seconds be the time that elapses after the car passes A .

In a **third** model, the car is assumed to move with constant speed 25 m s^{-1} from A until time $t = 0.2$, the car then decelerates uniformly at 6 m s^{-2} until it reaches a speed of 13 m s^{-1} . The car then maintains this speed until it reaches the road sign.

- (c) Use this third model to find the value of t at which the car reaches the sign.

(4)

DR ESAM AHMED

TOPIC 04 | FAMILY 04

Rebound and repeated-impact kinematics

WHAT THIS FAMILY ASKS

Track vertical flights separated by instantaneous impacts that change the velocity.

VARIANT ROUTE

V01 Repeated rebounds or speed ratio

V02 Rebound linked to impulse

V03 Time above a height across rebounds

3 classified questions

29 total marks

2. A small ball is released from rest from a point that is 40m above horizontal ground. The ball bounces on the ground and rebounds vertically. Each time the ball bounces on the ground, the speed of the ball is instantaneously reduced by 50%. The ball is modelled as a particle moving freely under gravity, from the instant when it is released until it first hits the ground, and between each successive bounce.
- (a) Find the time from the instant when the ball is released from rest to the instant when it hits the ground for the second time.

(5)

DR ESAM AHMED

2. A small ball is released from rest from a point that is 40m above horizontal ground. The ball bounces on the ground and rebounds vertically. Each time the ball bounces on the ground, the speed of the ball is instantaneously reduced by 50%. The ball is modelled as a particle moving freely under gravity, from the instant when it is released until it first hits the ground, and between each successive bounce.
- (b) Find the maximum height reached by the ball above the ground after the ball's third bounce.

(4)

DR ESAM AHMED

- 7 At time $t = 0$, a small ball A is projected vertically upwards with speed 8 m s^{-1} from a fixed point on horizontal ground.
The ball hits the ground again for the first time at time $t = T_1$ seconds.

Ball A is modelled as a particle moving freely under gravity.

- (a) Show that $T_1 = 1.63$ to 3 significant figures.

(2)

DR ESAM AHMED

- 7 At time $t = 0$, a small ball A is projected vertically upwards with speed 8 m s^{-1} from a fixed point on horizontal ground.

The ball hits the ground again for the first time at time $t = T_1$ seconds.

Ball A is modelled as a particle moving freely under gravity.

After the first impact with the ground, A rebounds to a height of 2 m above the ground.

Given that the mass of A is 0.1 kg,

- (b) find the magnitude of the impulse received by A as a result of its first impact with the ground.

(5)

DR ESAM AHMED

- 7 At time $t = 0$, a small ball A is projected vertically upwards with speed 8 m s^{-1} from a fixed point on horizontal ground.
The ball hits the ground again for the first time at time $t = T_1$ seconds.

Ball A is modelled as a particle moving freely under gravity.

At time $t = 1$ second, another small ball B is projected vertically upwards from another point on the ground with speed 5 m s^{-1}

Ball B is modelled as a particle moving freely under gravity.

At time $t = T_2$ seconds ($T_2 > 1$), A and B are at the same height above the ground for the first time.

- (c) Find the value of T_2

(4)

DR ESAM AHMED

4. The point O is 28.4m above a horizontal plane. A small ball is projected vertically upwards from O with speed $U \text{ m s}^{-1}$ and first hits the plane with speed 24 m s^{-1}

Air resistance is assumed to be negligible.

- (a) Find the value of U .

(3)

DR ESAM AHMED

4. The point O is 28.4m above a horizontal plane. A small ball is projected vertically upwards from O with speed $U \text{ m s}^{-1}$ and first hits the plane with speed 24 m s^{-1}

Air resistance is assumed to be negligible.

Each time the ball hits the plane, the speed of the ball is instantaneously halved.

- (b) Find the **total** time, **after the ball has hit the plane for the first time**, for which the ball is at least 4m above the plane.

(6)

DR ESAM AHMED

TOPIC 04 | FAMILY 05

Meeting of constant-acceleration particles

WHAT THIS FAMILY ASKS

Find when two scalar position functions are equal and extract event velocities.

VARIANT ROUTE

V01 Vertical projected-particle meeting

2 classified questions

20 total marks

3. A particle, P , is projected vertically upwards with speed U from a fixed point O . At the instant when P reaches its greatest height H above O , a second particle, Q , is projected with speed $\frac{1}{2}U$ vertically upwards from O .

(a) Find H in terms of U and g .

(2)

DR ESAM AHMED

3. A particle, P , is projected vertically upwards with speed U from a fixed point O . At the instant when P reaches its greatest height H above O , a second particle, Q , is projected with speed $\frac{1}{2}U$ vertically upwards from O .
- (b) Find, in terms of U and g , the time between the instant when Q is projected and the instant when the two particles collide.

(6)

DR ESAM AHMED

3. A particle, P , is projected vertically upwards with speed U from a fixed point O . At the instant when P reaches its greatest height H above O , a second particle, Q , is projected with speed $\frac{1}{2}U$ vertically upwards from O .

(c) Find where the two particles collide.

(3)

DR ESLAM AHMED

4. The point A is 10 m above horizontal ground.
At time $t = 0$, a particle P is projected vertically upwards with speed 5 m s^{-1} from A .
Particle P moves freely under gravity.

(a) Find the greatest height above A reached by P .

(3)

DR ESAM AHMED

4. The point A is 10 m above horizontal ground.
At time $t = 0$, a particle P is projected vertically upwards with speed 5 m s^{-1} from A .
Particle P moves freely under gravity.

The point B is on the ground, vertically below A .
At time $t = 1$ second, a particle Q is projected vertically upwards with speed 7 m s^{-1} from B .

Particle Q moves freely under gravity.

Particles P and Q collide at time $t = T$ seconds.

- (b) Find the value of T .

(4)

DR ESAM AHMED

4. The point A is 10 m above horizontal ground.
At time $t = 0$, a particle P is projected vertically upwards with speed 5 m s^{-1} from A .
Particle P moves freely under gravity.

The point B is on the ground, vertically below A .
At time $t = 1$ second, a particle Q is projected vertically upwards with speed 7 m s^{-1} from B .
Particle Q moves freely under gravity.

Particles P and Q collide at time $t = T$ seconds.

- (c) Find the speed of P at the instant immediately before the particles collide.

(2)

DR ESAM AHMED